

Visualizing Genomic Features for Disease Subtype Cluster Analysis

A Machine Learning approach using PCA, t-SNE & clustering to turn thousands of gene activity levels into clear 2D/3D disease-subtype maps.

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Genetic data is huge, noisy, and **hard to read**.

Modern healthcare generates thousands of gene-activity measurements per patient — but raw genomic data is too complex for doctors to interpret directly.



Massive dimensionality

01

Each patient sample carries thousands of gene features, overwhelming standard analysis tools.



Hidden background noise

02

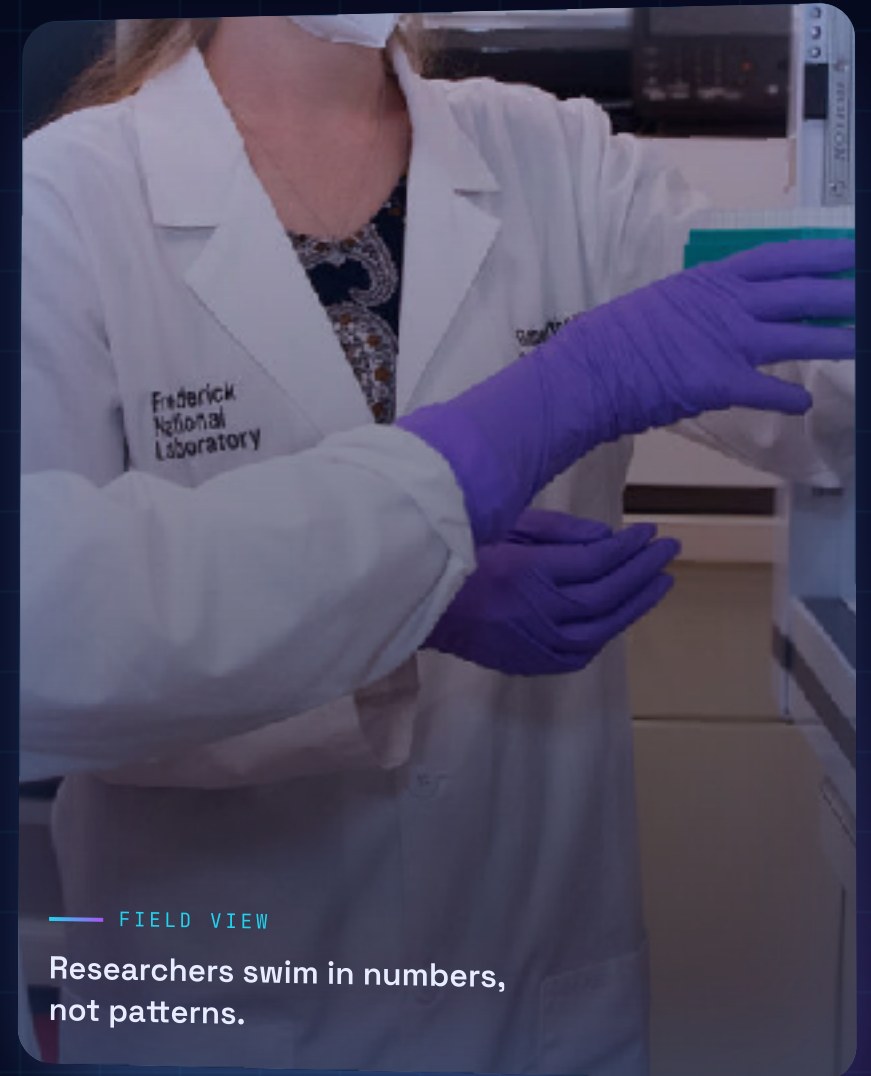
Random variation buries the real biological signal that distinguishes disease subtypes.



Patterns you cannot see

03

Similar patient groups stay invisible in raw numeric tables — clusters need visualization.



FIELD VIEW

Researchers swim in numbers,
not patterns.

Turn thousands of gene features into clear, visual **disease-subtype maps**.

Build a step-by-step machine-learning pipeline that simplifies complex genomic data, removes background noise, and groups similar patients into distinct disease subtypes — so doctors can see the patterns, not just the numbers.



01

Simplify

Reduce thousands of features to 2D / 3D visual maps.



02

Clarify

Strip noise without losing biological detail.



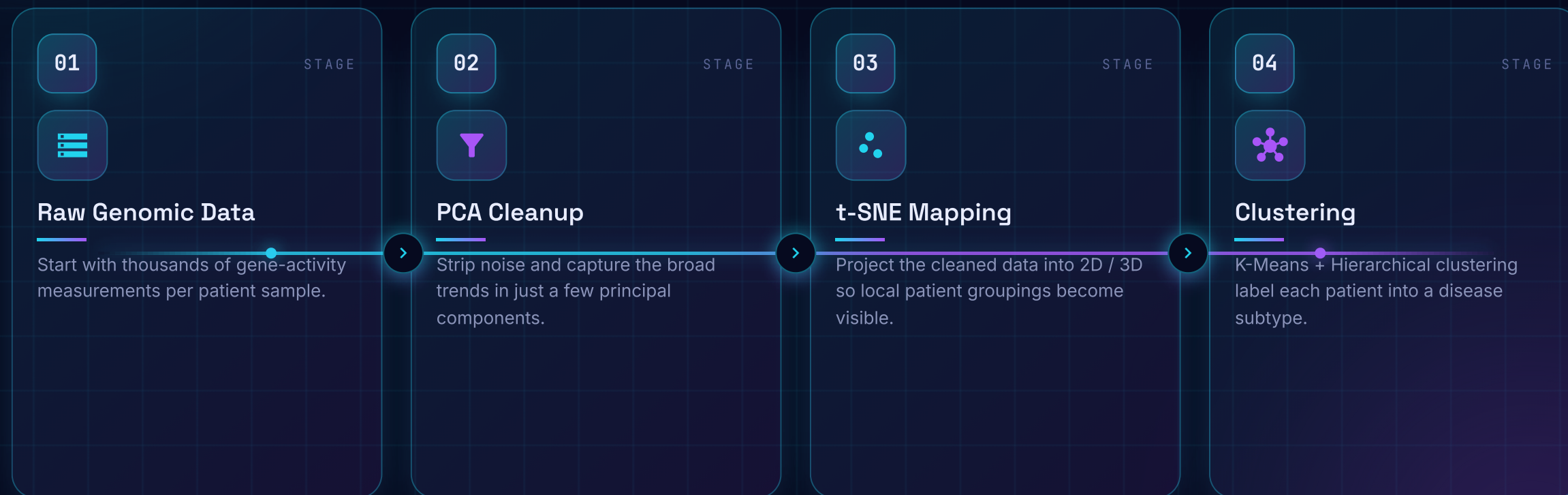
03

Group

Cluster patients into clear disease subtypes.

A 4-stage pipeline: from raw genes to clear disease subtypes.

Each stage hands a cleaner, smaller dataset to the next — making the final clustering both faster and more accurate.



FLOW ~20K genes → 10-50 PCs → 2D / 3D map → subtype labels

Each stage reduces dimensionality & sharpens signal

Principal Component Analysis

Summarizes **broad trends** across thousands of genes.

Think of PCA as a summarizer. It compresses thousands of correlated gene measurements into a handful of new variables called **principal components** — each one capturing the biggest remaining pattern in the data.

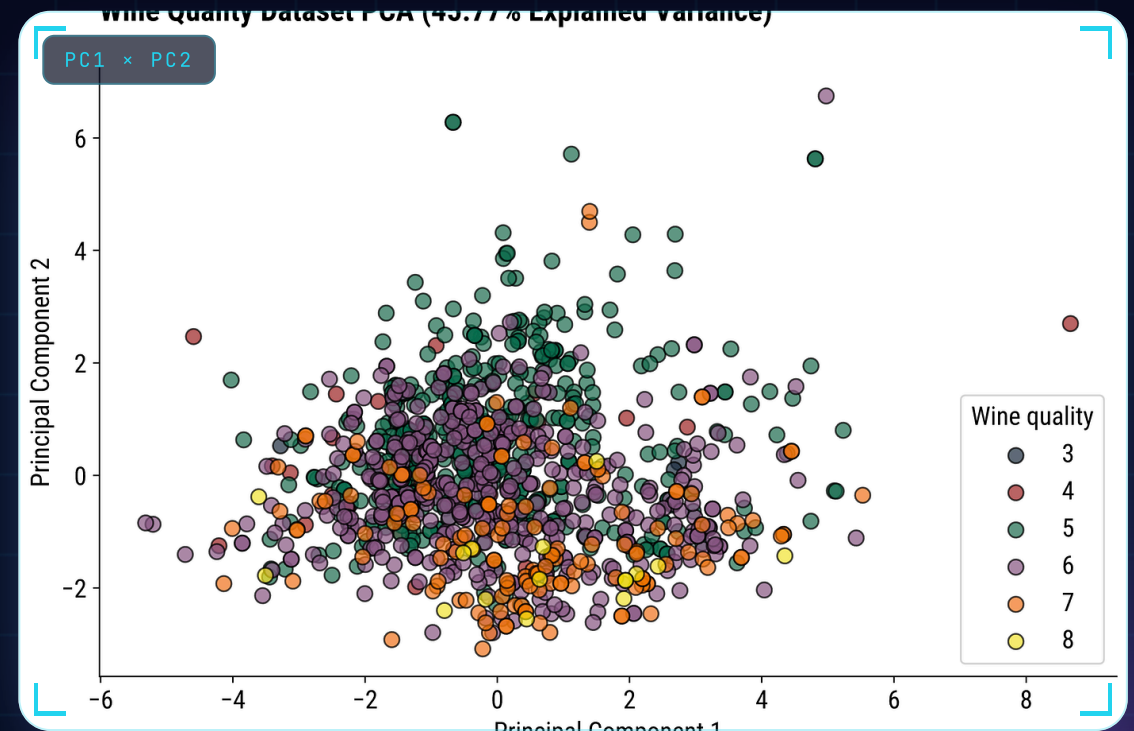
We keep only the top components (covering most of the variance) and discard the rest as noise. The result: a much smaller, cleaner dataset that still preserves the dominant biological signals.

INPUT • ~20,000 GENES

OUTPUT • 10-50 COMPONENTS

NOISE REMOVED • 70-90%

• RAW DATA → • PCA CLEANUP → • T-SNE



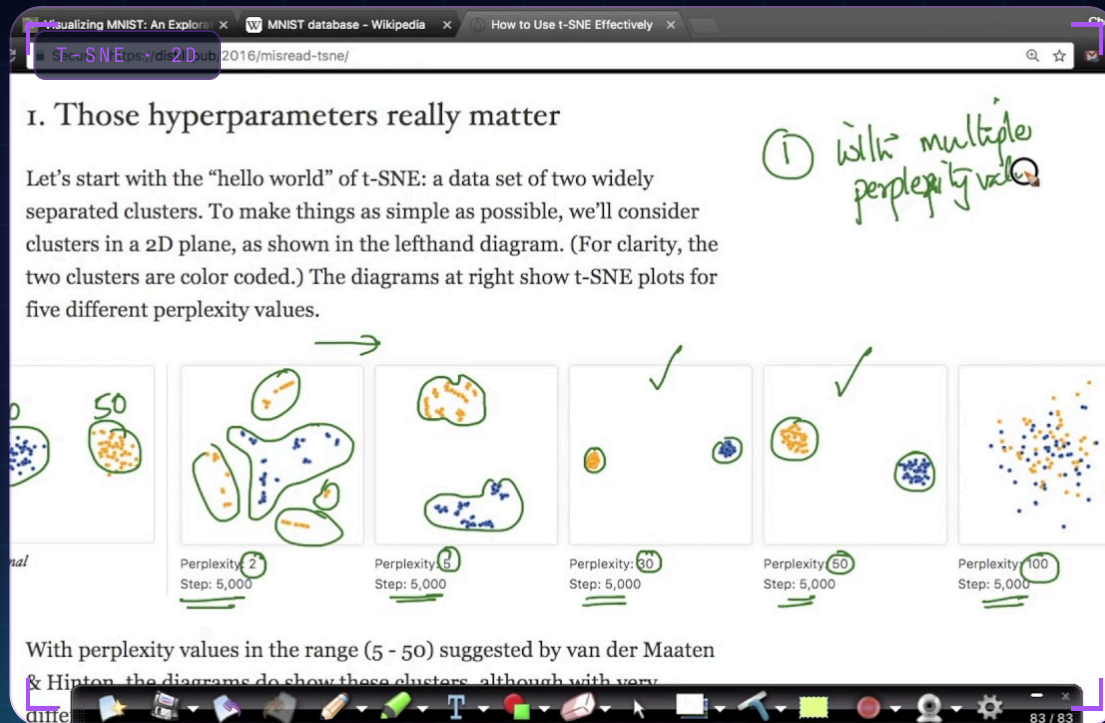
PCA scatter — broad structure becomes visible

t-Distributed Stochastic Neighbor Embedding

Reveals the **local detail** PCA cannot.

PCA captures the big picture. **t-SNE zooms in**. It places each patient on a 2D or 3D canvas so that samples with similar gene activity stay close together — and dissimilar samples push apart.

The output looks like a star map. Each dot is a patient, and tight clusters often correspond to a real disease subtype hiding inside the original data.



INPUT · PCA-REDUCED COMPONENTS

OUTPUT · 2D / 3D PATIENT MAP

BEST FOR · LOCAL STRUCTURE

● RAW → ● PCA → ● **T-SNE MAP** → ● CLUSTER

● t-SNE map — tight, separated patient clusters

K-Means + Hierarchical Clustering — turn the visual map into **labeled subtypes**.

Once patients sit in clear groups on the t-SNE map, clustering algorithms assign each one a subtype label — no manual inspection needed.



ALGORITHM A

CENTROID-BASED

K-Means Clustering

Fast, centroid-based grouping.

- Picks **K** cluster centers, assigns patients to nearest center
- Iterates until group centers stabilize
- Best when subtypes are roughly spherical and similar in size



ALGORITHM B

TREE-BASED

Hierarchical Clustering

Builds a tree of patient similarity.

- Merges closest patient pairs step-by-step into a **dendrogram**
- Cut the tree at the chosen depth to get **K** subtypes
- Best when subtype hierarchy / nesting matters

K-MEANS LABELS ON T-SNE MAP

0

Each color = one predicted **disease subtype**

A B C D

Tested on **real-world genomic data**.

We use the TCGA Gene Expression dataset as our primary source, with two well-known public alternatives considered for cross-validation and extension.

⚙️ PRIMARY



TCGA — The Cancer Genome Atlas

2.5 petabytes of tumor + matched-normal data covering 11,000+ patients across 33 cancer types. The gold standard for molecular subtype discovery.

11,000+ PATIENTS

33 CANCER TYPES

2.5 PB TOTAL



33 CANCER TYPES · MUTATIONAL LANDSCAPE

🔗 Source: NCI / cancer.gov · GDC Portal

📁 ALT · 01



GDC — Genomic Data Commons

Standardized pipeline + harmonized clinical metadata for TCGA and beyond.

- Unified RNA-seq pipeline across studies
- Direct clinical / subtype annotations available

📁 ALT · 02



GTEx — Genotype-Tissue Expression

Healthy-tissue gene expression from 948 donors across 54 tissues.

- Perfect normal-tissue baseline to compare tumors against
- Often combined with TCGA to filter cancer-specific signal

📄 GEO (Gene Expression Omnibus) was also surveyed as a complementary repository for validation cohorts.

Cleaner data, sharper subtypes, faster runs.

The PCA-then-t-SNE pipeline measurably improves subtype separation, clustering quality, and processing speed versus running clustering directly on raw gene-expression data.

METRIC · 01



↑ Sharp

Visual Clarity

Disease subtypes separate into tighter, more obvious clusters on the 2D / 3D map.

VS. RAW-DATA BASELINE

METRIC · 02



↑ Better

Grouping Accuracy

Clustering on the reduced data agrees more strongly with known clinical subtype labels.

SILHOUETTE · ARI

METRIC · 03



↓ Faster

Processing Time

PCA pre-shrinks the data, so t-SNE and K-Means converge in a fraction of the raw-data runtime.

END-TO-END RUNTIME



A visual tool for precision medicine.

Doctors and researchers gain a clear visual lens to **discover new disease categories**, **confirm existing subtypes**, and **tailor treatments** to each patient group — turning thousands of invisible measurements into actionable insight.

CLINICAL IMPACT

Actionable Insight

Thank You.

Questions & discussion welcome.

👁 Visualize

✨ Simplify

👥 Cluster



2026-27

PROJECT CREDITS

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Department of Computer Science & Technology · A.Y. 2026-27